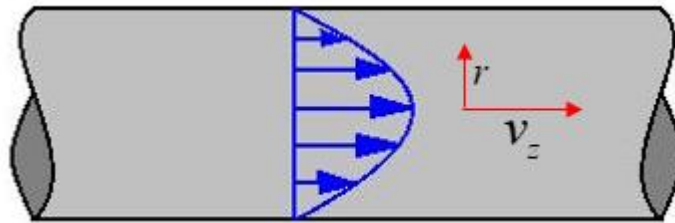


Practice Assignment -1 (Transport Phenomena), Spring-2023

(No submission is required, this is for practice only)

Consider a pressure driven unidirectional flow of a fluid (density = ρ , viscosity = μ) through a cylindrical pipe (radius = R , length = L), as shown in figure below. Consider a linear pressure gradient along the flow, P_i and P_0 are inlet and outlet pressure respectively.



We all know that the steady state profile is parabolic, however at early times (starting from $t=0$), it is not parabolic, in fact velocity profile evolves and eventually becomes parabolic. This phenomenon of temporal evolution of flow profile is known as *Start-up of Laminar Flow*. In this problem we are going to capture such evolution of velocity profile. Assume flow is fully developed at any instant of time, so that there is no variation in velocity-magnitude in axial direction (z), there exists only radial gradient of velocity.

Perform the following steps:

a) Write down the governing equation, initial condition and boundary conditions in order to obtain velocity profile: $v_z(r, t)$. (use cylindrical-coordinate system)

b) Consider the characteristic length as R , using order of magnitude analysis, show that characteristic velocity scale (U) and characteristic time scale (T) are given by,

$$U = \frac{(P_i - P_0)R^2}{\mu L}, \quad T = \frac{R^2}{\mu / \rho} \text{ respectively.}$$

c) Using the above characteristic values, non-dimensionalise the governing equation to obtain the equation in terms of $\phi\left(\eta = \frac{v_z}{U}\right)$, $\eta\left(\eta = \frac{r}{R}\right)$, and $\tau\left(\tau = \frac{t}{T}\right)$.

d) In order to obtain non-dimensional velocity ϕ , we propose that $\phi = \phi_s + \phi_t$, where ϕ_s and ϕ_t are steady state and unsteady component respectively. Obtain steady state profile ϕ_s in terms of η .

e) Solve the non-dimensional governing equation (obtained in part c) **numerically or analytically**, and plot $\phi(\eta, \tau)$ for $\tau = 0.005a, 0.015a, 0.1a, a$, where a is last two digits of your roll number. For roll number ending with 00, consider $a=12$. Consider τ values up to 4 decimal places. You are free to choose any value of physical properties (density, pressure difference, viscosity), geometrical dimension.

Note: In case you are solving numerically, kindly upload your Matlab/python script file.

Hint: Out of two boundary conditions (BC), one will involve axis-symmetric BC. While solving analytically use separation of variables, for special component you will get famous Bessel differential equation (https://en.wikipedia.org/wiki/Bessel_function). Hence your final answer should be in terms of Bessel functions.